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Study on non-convexity for Zhang-Zhu strength criterion based on microfracture mechanics

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ABSTRACT

The Zhang-Zhu (ZZ) strength criterion, as a three-dimensional Hoek-Brown strength criterion, accurately characterizes the deep rock failure under the true triaxial stress condition ($\sigma_1>\sigma_2>\sigma_3$). However, its failure envelope in the deviatoric plane shows non-smoothness and non-convexity, with the underlying physical mechanism for the non-convexity requiring further exploration. This study derives a micro-Zhang-Zhu (micro-ZZ) strength criterion from the microfracture mechanics, following the formulation of the ZZ strength criterion. The validation against the true triaxial test reveals that 87.2% of the data points fall within a ±5% relative error, indicating that fracture behavior from microcrack clusters is strongly consistent with true triaxial test results. The micro-ZZ strength criterion shows a maximum mean absolute percentage error (MAPE) of only 2.39% relative to the ZZ strength criterion, confirming its high consistency with the ZZ strength criterion. Therefore, a correlation between microfracture mechanisms and macroscopic non-convexity of the Zhang-Zhu strength criterion is established based on the micro-ZZ strength criterion. The failure envelopes of both strength criteria closely match, exhibiting strong agreement in their intersection angles θ_d. The effect of the microcrack density on the non-convexity is further explored to reveal the underlying physical mechanism of non-convexity. The results indicate that the non-convexity increases with the microcrack density and approaches 90° as the density tends to zero, indicating the microcrack is one of the micromechanical mechanisms contributing to the non-convexity. Finally, the effects of parameters on the non-convexity and the sensitivity of these parameters are discussed. Five stress state and material parameters influence the non-convexity, with the friction coefficient μ and the compression-extension stress ratio σ_c/σ_t identified as the dominant parameters, with the strongest interaction between them.

Nomenclature

σ_1, σ_2, σ_3	Major, intermediate, and minor principal stress
σ_c	Unconfined compression strength
m_i	Rock parameter
m_b, s and a	Rock mass parameters
τ_{oct}	Octahedral shear stress
σ_m	Mean stress
$\sigma_{m,2}$	Effective mean stress
J_2	Second deviatoric stress invariant
J_1	First deviatoric stress invariant
I_1	First principal stress invariant
φ	Rotation angle of penny-shaped microcrack to σ_2-axis
θ	Dip angle of penny-shaped microcrack
l	Radius of penny-shaped microcrack

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σ_{11}', σ_{12}', σ_{13}'	Normal, tearing shear, and sliding shear stresses of penny-shaped microcrack
K_I, K_{II}, K_{III}	Stress intensity factors for fracture Mode I, II, and III
K_{IC}, K_{IIC}, K_{IIIC}	Fracture toughness values for Mode I, II, and III
σ_t	Uniaxial tensile strength
κ	Fracture coefficient
μ	Friction coefficient
ν	Poisson's ratio
β	Variable for fracture parameters: $\beta = \kappa\sigma_t(2-\nu)/2$
α	Micro-failure orientation angle
φ_a	Actual rotation angle of microcracks
σ_{1a}	Local strength corresponding to actual rotation angle of microcracks
φ_m	Most favorable orientation of penny-shaped microcrack cluster

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C_1, ξ	Variables determined by rock material: $C_1 = 0.0035m_i - 0.40$, $\xi = \left(-1-\sqrt{3}\tan\theta_\sigma\right)\cdot(0.0035m_i - 0.40)/2$
γ	Lode parameter
θ_σ	Lode angle
MAPE	Mean absolute percentage error
RE	Relative error between theoretical and tested values
θ_d	Intersection angle
l_{1d}, k_{1d}	Radius of failure envelope and its slope
l_{2d}, k_{2d}	Tangent line of failure envelope and its slope
θ_{zd}, θ_{md}	Intersection angle θ_d for Zhang-Zhu strength criterion and micro-Zhang-Zhu strength criterion
θ_{TE}	Non-convex corner
$\theta_{z\mathrm{TE}}$, $\theta_{m\mathrm{TE}}$	Non-convex corners for Zhang-Zhu strength criterion and micro-Zhang-Zhu strength criterion
S_i	First-order sensitivity index
S_{ij}	Second-order sensitivity index
ST_i	Total-order sensitivity index

1. Introduction

With the advancement of underground engineering into greater depths, geological conditions become increasingly complex, and deep rocks are commonly subjected to the true triaxial stress condition ($\sigma_1>\sigma_2>\sigma_3$).[1–3] Despite its widespread use over the last forty years and its ability to reflect the nonlinear behavior of rocks,[4–6] the Hoek-Brown (HB) criterion neglects the influence of the intermediate principal stress σ_2, limiting its accuracy in predicting the deep rock strength at true triaxial stress conditions. To incorporate the influence of σ_2, several three-dimensional (3D) HB strength criteria were developed.[7–12] Among them, the Zhang-Zhu (ZZ) strength criterion[9,10] based on the HB and Mogi[13] strength criteria demonstrates high accuracy for true triaxial test results. As it reduces to the original HB formulation at triaxial compression (TC) and triaxial extension (TE) states, it can be regarded as a true 3D HB strength criterion and directly employs the same rock parameters.

The ZZ strength criterion inherits the characteristic of the Mogi[13] strength criterion, exhibiting a non-smooth and non-convex failure envelope in the deviatoric plane, which contradicts the Drucker postulate of convexity.[14] This inherent non-convexity may lead to numerical singularities and convergence issues in computational implementations.[15–17] To address the problem, Zhang et al.[18] introduced three Lode dependences into the generalized ZZ strength criterion to ensure smoothness and convexity. Moreover, the mean stress term in the expression was also reconstructed to satisfy these requirements partially.[15,19] However, these modifications resolve the non-convexity by adjusting the mathematical expression, while the underlying physical mechanism of the non-convexity for the ZZ strength criterion requires further study.

The ZZ strength criterion, developed based on two empirical strength criteria, primarily emphasizes the macroscopic failure behavior of the rock and the empirical summary of experimental data. However, as a discontinuous medium containing numerous microcracks, rock failure is inherently governed by the initiation and propagation of these microcracks. Griffith[20] suggested that brittle materials fracture when the energy released by crack growth surpasses the energy required to create new crack surfaces. Wiebols and Cook[21] indicated that rock failure initiates once the additional strain energy concentrated around microcracks surpasses a critical limit. In addition to representing rock failure through microcrack energy, microfracture models can be constructed to explore rock failure from microfracture mechanics. Zuo et al.[22,23] constructed the microcrack models for rock failures and theoretically derived the HB strength criterion, which was extended into generalized nonlinear 3D forms.[24,25] However, the existing 3D HB strength criteria based on the microfracture mechanics ensure consistency with the HB strength criterion only at the TC state, whereas the ZZ strength criterion maintains consistency at both TC and TE states. Therefore, it is attempted to derive the ZZ strength criterion based on microfracture mechanics, providing insight into its non-convexity within the deviatoric plane.

In this study, a micro-Zhang-Zhu (micro-ZZ) strength criterion is derived from the microfracture mechanics, following the formulation of the ZZ strength criterion. Its accuracy is validated by true triaxial test results and theoretical comparison with the ZZ strength criterion. The non-convexity for the ZZ strength criterion in the deviatoric plane is interpreted through the micro-ZZ strength, and the underlying physical mechanism is revealed by exploring the effect of microcrack density within the rock on the non-convexity. Finally, the effects and sensitivities of stress state and material parameters on the non-convexity are discussed to identify the key parameters.

2. Formulation of ZZ strength criterion and its intrinsic nonconvexity

2.1. Formulation of ZZ strength criterion

As a 3D HB strength criterion recommended by the International Society for Rock Mechanics and Rock Engineering (ISRM),[11] the ZZ strength criterion integrates the Mogi[13] criterion into the HB strength criterion, with its formulation process detailed accordingly.

Based on extensive triaxial tests on intact rock specimens, Hoek and Brown[4,5] proposed the HB strength criterion, which remains the most extensively adopted empirical criterion for rock strength assessment.

$$\sigma_1 = \sigma_3 + \sigma_c\left(m_i\frac{\sigma_3}{\sigma_c}+1\right)^{0.5} \tag{1}$$

where σ_1 and σ_3 represent the major and minor principal stresses, respectively; m_i is the material constant for rock, defined as the rock parameter; σ_c denotes the unconfined compression strength.

To extend the applicability of the HB strength criterion to rock masses, Hoek et al.[6] proposed the generalized HB strength criterion, formulated as

$$\sigma_1 = \sigma_3 + \sigma_c\left(m_b\frac{\sigma_3}{\sigma_c}+s\right)^{a} \tag{2}$$

where m_b, s, and a are rock mass parameters that reflect the properties of rock masses.

The HB strength criterion neglects the influence of σ_2, though extensive studies demonstrating its significant effect on rock strength.[13,26–31] Mogi[13] first designed a true triaxial testing machine and conducted extensive tests on different rocks to investigate the effect of σ_2 on rock strength. Based on the true triaxial test data, Mogi[13] established a relationship between the octahedral shear stress τ_{oct} and the effective mean stress $\sigma_{m,2}$ rather than the mean stress σ_m, which is expressed as

$$\tau_{oct} = f\left(\sigma_{m,2}\right) \tag{3}$$

where $f(\,)$ represents an increasing function. τ_{oct}, $\sigma_{m,2}$, and σ_m denote the octahedral shear stress, effective mean stress, and mean stress, respectively, and defined by

$$\tau_{oct} = \frac{1}{3}\sqrt{\left(\sigma_1-\sigma_2\right)^2+\left(\sigma_2-\sigma_3\right)^2+\left(\sigma_3-\sigma_1\right)^2} \tag{4}$$

$$\sigma_{m,2} = \frac{\sigma_1+\sigma_3}{2} \tag{5}$$

$$\sigma_m = \frac{\sigma_1+\sigma_2+\sigma_3}{3} \tag{6}$$

Combining the HB and Mogi[13] criterion, Zhang and Zhu[9] proposed the ZZ strength criterion for rocks.

$$\frac{9}{2\sigma_c}\tau_{oct}^2+\frac{3}{2\sqrt{2}}m_b\tau_{oct}-m_b\sigma_{m,2}-s\sigma_c = 0 \tag{7}$$

To make it applicable to rock masses, Zhang[10] developed a generalized ZZ (GZZ) strength criterion, which is expressed as

$$\frac{1}{\sigma_c^{(1/a-1)}}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right)^{1/a}+\frac{m_b}{2}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right)-m_b\sigma_{m,2}-s\sigma_c=0 \tag{8}$$

For intact rocks (where $m_b = m_i$, $s = 1$, and $a = 0.5$), the GZZ strength criterion simplifies to its original version. Since the ZZ strength criterion reduces to the HB strength criterion at TC and TE states, it can be regarded as a true 3D HB strength criterion. This allows the ZZ strength criterion to directly adopt the identical parameters of the HB strength criterion that have been accumulated through testing and engineering experience.

2.2. Non-convexity for ZZ strength criterion in deviatoric plane

The ZZ strength criterion shows intrinsic non-convexity in the deviatoric plane, as discovered by comparing it with other 3D HB strength criteria. For example, the Pan-Hudson strength criterion is developed by Pan and Hudson.[7]

$$\frac{9}{2\sigma_c}\tau_{oct}^2+\frac{3}{2\sqrt{2}}m_b\tau_{oct}-m_b\sigma_m-s\sigma_c=0 \tag{9}$$

Priest[8] proposed the Priest strength criterion, which is expressed as follows:

$$\sqrt{J_2}=AJ_1+B \tag{10}$$

$$J_2=\frac{1}{6}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right] \tag{11}$$

$$J_1=\sigma_m=\frac{\sigma_1+\sigma_2+\sigma_3}{3} \tag{12}$$

where A and B are empirical parameters[8]; J_1 and J_2 denote the first and second deviatoric stress invariants.

Fig. 1 illustrates the failure envelopes of the 3D HB strength criteria in the deviatoric plane. The failure envelope of the HB strength criterion is composed of six parabolic curves with slight curvature, which is only meaningful at TC and TE states. The Pan-Hudson and Priest strength criteria exhibit circular failure envelopes that meet the requirements of smoothness and convexity. The strength determined by the ZZ strength criterion is equivalent to the HB strength criterion at both TC and TE states. Different from the circular failure envelope, the ZZ strength criterion shows a non-smooth and non-convex curvilinear hexagonal envelope, which is a typical manifestation of the Mogi-type strength criteria.[13] In previous studies, such non-convexity was modified to enforce convexity, which may obscure the intrinsic weakening regions and overestimate the rock strength along specific stress paths. Explicit consideration of non-convexity provides a more reliable assessment of rock failure under complex stress states, particularly in deep underground engineering and high σ_2/σ_3 conditions approaching the TE state, where sudden engineering disasters often occur.[32] Moreover, the true triaxial test data within the same deviatoric plane obtained by Su et al.[32] indicate the inherent non-convexity of rock failure behavior, highlighting the necessity to explore the underlying physical mechanisms of the non-convexity.

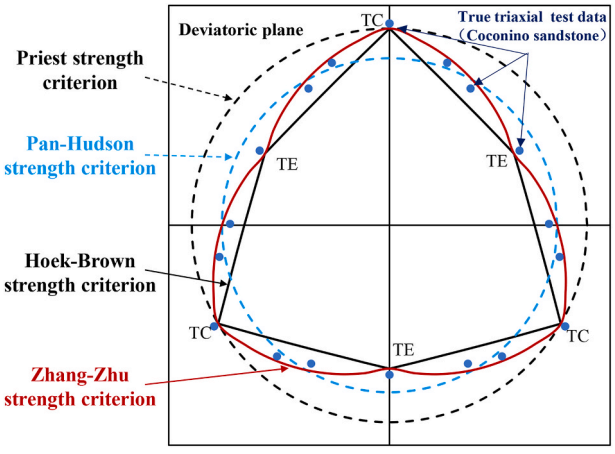


Fig. 1. Deviatoric plane of four strength criteria at $I_1 = 740$ MPa for Coconino sandstone[33] with $\sigma_c = 56.1$ MPa, $m_i = 21$, $s = 1$, and $a = 0.5$.

Based on the true triaxial test data, the cause of the non-convexity for the ZZ strength criterion is preliminarily analyzed. Since the lack of true triaxial test data at the same mean stress,[28,33,34] the data points with the first principal stress invariant I_1 within ±5% discrepancy are considered as representing the same mean stress. The data points for the Coconino sandstone with I_1 equals 740 MPa are represented in Fig. 1. These data points are distributed around the failure envelope of the ZZ strength criterion, indicating that the failure envelope of rocks has a similar non-convexity. This implies that the non-convexity for the ZZ strength criterion is an intrinsic characteristic of rock failure.

3. Derivation of micro-ZZ strength criterion based on microfracture mechanics

The preceding analysis indicates that the non-convexity for the ZZ strength criterion in the deviatoric planes is intricately correlated to the rock failure mechanisms. Rock material, as a discontinuous medium containing numerous microcracks, its failure is inherently governed by the propagation of internal microcracks. Therefore, a micro-Zhang-Zhu (micro-ZZ) strength criterion is derived from microfracture mechanics, following the formulation of the ZZ strength criterion, to explore the physical mechanism of the non-convexity for the ZZ strength criterion.

3.1. 3D microcrack model for rock failure

Numerous microcracks have been identified within rock materials,[20,35–39] with common models including the Griffith microcracks[20,35] and the penny-shaped microcracks,[36] among others. It is worth noting that the natural microcracks in rocks are generally anisotropic, which may affect the energy release rate, crack propagation direction, and macroscopic failure behavior. However, the ZZ strength criterion was developed for isotropic rocks and restricted to isotropic behavior.[9,10] To maintain theoretical consistency with the original ZZ strength criterion and to avoid introducing unnecessary complexity, the penny-shaped microcrack model under the assumption of isotropy is adopted, and three fundamental hypotheses are proposed:[22–25,40] (1) rock material contains a random distribution of numerous penny-shaped microcracks; (2) microcracks are sparsely distributed, interactions between the microcracks are neglected[25]; (3) the matrix material surrounding the microcracks is isotropic and elastic.

Based on the fundamental hypotheses, a single penny-shaped microcrack in an isotropic medium subjected to uniform far-field loading is constructed, as illustrated in Fig. 2. The global coordinate system (O-$\sigma_1\sigma_2\sigma_3$) and the local coordinate system (O-$\sigma_{11}'\sigma_{12}'\sigma_{13}'$) are established. φ represents the rotation angle of the penny-shaped microcrack to the σ_2-axis, while θ denotes its dip angle. The stresses within the local coordinate system are calculated as follows[41]:

$$\sigma_{ij}' = g_{ik}g_{jl}\sigma_{kl} \tag{13}$$

where g_{ij} is the transformation matrix defined as:

$$g_{ij}=\begin{bmatrix}\cos\theta & \sin\theta\sin\varphi & -\sin\theta\cos\varphi\\ 0 & \cos\varphi & \sin\varphi\\ \sin\theta & -\cos\theta\sin\varphi & \cos\theta\cos\varphi\end{bmatrix} \tag{14}$$

σ_{11}' is the normal stress of the penny-shaped microcrack, σ_{12}' and σ_{13}'

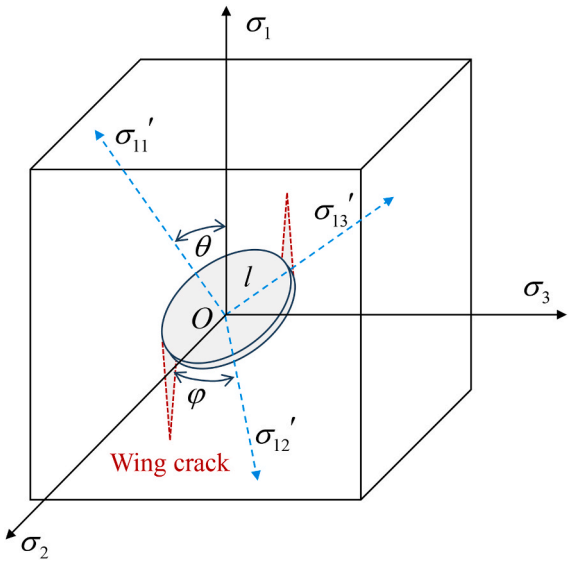


Fig. 2. Single penny-shaped microcrack model for rock failure.

represent the tearing and sliding shear stresses, respectively. They are calculated as follows:

$$\begin{cases} \sigma_{11}{}' = \sigma_1 \cos^2\theta + \sigma_2 \sin^2\theta \sin^2\varphi + \sigma_3 \sin^2\theta \cos^2\varphi \\ \sigma_{12}{}' = \sigma_2 \sin\theta \sin\varphi \cos\varphi - \sigma_3 \sin\theta \sin\varphi \cos\varphi \\ \sigma_{13}{}' = \sigma_1 \cos\theta \sin\theta - \sigma_2 \cos\theta \sin\theta \sin^2\varphi - \sigma_3 \cos\theta \sin\theta \cos^2\varphi \end{cases} \tag{15}$$

Under the mixed fracture mode, the propagation for the penny-shaped microcrack is expressed as follows[42–45]:

$$\frac{K_{\mathrm{I}}}{K_{\mathrm{IC}}} + \frac{K_{\mathrm{II}}}{K_{\mathrm{IIC}}} + \frac{K_{\mathrm{III}}}{K_{\mathrm{IIIC}}} = 1 \tag{16}$$

where K_{I}, K_{II}, and K_{III} denote the stress intensity factors for Mode I, II, and III, respectively, while the K_{IC}, K_{IIC}, and K_{IIIC} represent the corresponding fracture toughness values, defined as the critical stress intensity factors for each fracture mode. Among them, the K_{IC} is determined as[23–25,43]

$$K_{\mathrm{IC}} = 2\sigma_t \sqrt{\frac{l}{\pi}} \tag{17}$$

where σ_t denotes the uniaxial tensile strength of rock; l denotes the radius of the penny-shaped microcrack.

Under true triaxial compression, all principal stresses remain compressive, leading to the stress intensity factor for Mode I (K_{I}) is equal to 0.[40,43] Therefore, crack deformation and propagation are dominated by frictional sliding, and the contribution of Mode I can be neglected. When the effective shear stress exceeds the friction resistance, the penny-shaped microcrack slides, which induces stress concentration on both sides of the microcrack and facilitates the formation of wing cracks.[46,47] The propagation of wing cracks is governed by the in-plane traction (sliding shear stress), whereas the out-plane traction (tearing shear stress) is neglected, rendering the contribution of Mode III insignificant.[48,49] Consequently, Eq. (16) is simplified into

$$K_{\mathrm{II}} = K_{\mathrm{IIC}} \tag{18}$$

The relationship between K_{IC} and K_{IIC} is expressed in a linear form.[42,50]

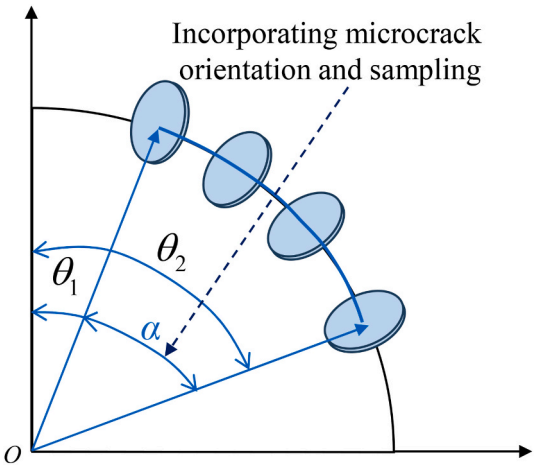


Fig. 3. Distribution zone of penny-shaped microcracks for a given rotation angle φ

$$K_{\mathrm{IIC}} = \kappa K_{\mathrm{IC}} \tag{19}$$

where κ represents the fracture coefficient.

According to Tada et al.,[51] the stress intensity factor K_{II} for the in-plane shear stress is determined by

$$K_{\mathrm{II}} = \frac{4(\sigma_{13}{}' - \mu\sigma_{11}{}')}{2-\nu}\sqrt{\frac{l}{\pi}} \tag{20}$$

where ν denotes the Poisson's ratio; μ is the friction coefficient.

Combining Eqs. (16)–(20), the propagation for the penny-shaped microcrack can be expressed as

$$\sigma_{13}{}' - \mu\sigma_{11}{}' = \beta \tag{21}$$

where β is defined as

$$\beta = \kappa\sigma_t \frac{2-\nu}{2} \tag{22}$$

3.2. Microfracture strength criterion based on single penny-shaped microcrack

Upon satisfying the propagation conditions, multiple wing cracks are initiated and propagated at the tips of the initial penny-shaped microcracks. By centralizing all the wing cracks, a sector distribution zone is obtained for a given rotation angle φ, as illustrated in Fig. 3. The dip angle within this distribution zone ranges from θ_1 to θ_2, with the center angle defined as the micro-failure orientation angle α, which characterizes the microcrack orientation and sampling.[23] Owing to the sparse distribution of microcracks in the adopted model, their orientations can be treated as statistically uniform. An increase in sampling density is equivalent to an expansion of the orientation range of microcracks participating in failure, as reflected by a larger α. Accordingly, the sensitivity of the numerical discretization is effectively evaluated through the parametric analysis of α.

Combining Eq. (15) and Eq. (21), the following equation is obtained.

$$\tan^2\theta(\beta + \mu\sin^2\varphi\sigma_2 + \mu\cos^2\varphi\sigma_3) + \tan\theta(\sin^2\varphi\sigma_2 + \cos^2\varphi\sigma_3 - \sigma_1) + (\beta + \mu\sigma_1) = 0 \tag{23}$$

Solving the above equation, the dip range of the penny-shaped microcracks (θ_1, θ_2) is determined by

$$\tan\theta_{1,2}=\frac{\sigma_1-\sin^2\varphi\sigma_2-\cos^2\varphi\sigma_3\mp\sqrt{\left(\sigma_1-\sin^2\varphi\sigma_2-\cos^2\varphi\sigma_3\right)^2-4(\beta+\mu\sigma_1)\left(\beta+\mu\sin^2\varphi\sigma_2+\mu\cos^2\varphi\sigma_3\right)}}{2\left(\beta+\mu\sin^2\varphi\sigma_2+\mu\cos^2\varphi\sigma_3\right)} \tag{24}$$

where θ_1 and θ_2 are the dip angles of the microcracks for a given φ, as illustrated in Fig. 3. Therefore, the α, defined as $\theta_2-\theta_1$, is calculated by

$$\tan\alpha=\frac{\sqrt{\left(\sigma_1-\sin^2\varphi\sigma_2-\cos^2\varphi\sigma_3\right)^2-4(\beta+\mu\sigma_1)\left(\beta+\mu\sin^2\varphi\sigma_2+\mu\cos^2\varphi\sigma_3\right)}}{2\beta+\mu\left(\sigma_1+\sin^2\varphi\sigma_2+\cos^2\varphi\sigma_3\right)} \tag{25}$$

The brittle failure in rock is complex, but it should be characterized by an invariant, referred to as the failure characteristic parameter, at the point of complete failure. Numerous tests have demonstrated that the brittle failure in rocks is not induced by the initiation of microcracks.[52–54] Instead, rock failure occurs once the micro-failure density exceeds the critical threshold. Consequently, the failure characteristic parameter is correlated to the micro-failure density, which is strongly influenced by the stress state[36,37] and is positively correlated to the α. Besides, the failure characteristic parameter should meet the following principles: (1) a simple mathematical expression; (2) a negative correlation between the confining pressure σ_3 and the α; (3) invariance of the failure characteristic parameter; (4) a high degree of agreement between theoretical results and tested results.

Given the complexity of the expressions for the α and $\tan(\alpha)$, the cosine of α is computed.

$$\cos\alpha=\frac{\mu}{\sqrt{1+\mu^2}}\left(1+2\frac{\sin^2\varphi\sigma_2+\cos^2\varphi\sigma_3+\beta/\mu}{\sigma_1-\sin^2\varphi\sigma_2-\cos^2\varphi\sigma_3}\right) \tag{26}$$

The $\cos(\alpha)$ is an increasing function of σ_3, indicating that α is negatively correlated with σ_3. Considering the influence of the stress state, the σ_2 and σ_3 are considered invariant, and the variation of $\cos(\alpha)$ with respect to σ_1 is then evaluated.

$$\left|\frac{\partial\cos\alpha}{\partial\sigma_1}\right|=2\frac{\mu}{\sqrt{1+\mu^2}}\frac{\sin^2\varphi\sigma_2+\cos^2\varphi\sigma_3+\beta/\mu}{\left(\sigma_1-\sin^2\varphi\sigma_2-\cos^2\varphi\sigma_3\right)^2} \tag{27}$$

Replacing σ_1 in Eq. (27) with σ_2, σ_3, and $\cos(\alpha)$, the following expression is obtained.

$$\left|\frac{\partial\cos\alpha}{\partial\sigma_1}\right|=\frac{\mu}{\sqrt{1+\mu^2}}\frac{\left(\sqrt{1+\mu^2}\,/\,\mu\cdot\cos\alpha-1\right)^2}{2\left(\sin^2\varphi\sigma_2+\cos^2\varphi\sigma_3+\beta/\mu\right)} \tag{28}$$

For constant σ_2, the increase of σ_3 results in a rise in the denominator of $|\partial\cos\alpha/\partial\sigma_1|$. Meanwhile, since higher σ_3 corresponds to larger $\cos(\alpha)$, the numerator of $|\partial\cos\alpha/\partial\sigma_1|$ also becomes larger. When the increasing multiples of the denominator and numerator are identical, $|\partial\cos\alpha/\partial\sigma_1|$ remains constant and can be regarded as a failure characteristic parameter.

For the uniaxial compression state, σ_1 equals σ_c, and both σ_2 and σ_3 are zero. The failure characteristic parameter is denoted as

$$\left|\frac{\partial\cos\alpha}{\partial\sigma_1}\right|=2\frac{\mu}{\sqrt{1+\mu^2}}\frac{\beta/\mu}{\sigma_c^2} \tag{29}$$

According to Eq. (27) and Eq. (29), the microfracture strength criterion is derived:

$$\sigma_1-\sin^2\varphi\sigma_2-\cos^2\varphi\sigma_3=\sqrt{\frac{\mu}{\beta}\sigma_c^2\left(\sin^2\varphi\sigma_2+\cos^2\varphi\sigma_3\right)+\sigma_c^2} \tag{30}$$

When applied to assess rock strength at the TC state ($\sigma_2=\sigma_3$), Eq. (30) is converted into Eq. (31), which closely resembles the form of the HB strength criterion. The m_i is equivalently represented by Eq. (32).

$$\sigma_1-\sigma_3=\sqrt{\frac{\mu}{\beta}\sigma_c^2\sigma_3+\sigma_c^2} \tag{31}$$

$$m_i=\frac{\mu\sigma_c}{\beta}=\frac{2\mu\sigma_c}{\kappa\sigma_t(2-\nu)} \tag{32}$$

3.3. Extension to microcrack cluster

It is worth noting that Eq. (30) is derived based on a single penny-shaped microcrack, but the rock strength is governed by the failure of the microcrack cluster. For determined σ_2 and σ_3, a larger rotation angle φ corresponds to a higher rock strength σ_1. When the actual rotation angle of the microcracks φ_a is less than φ, the local strength σ_{1a} is less than the theoretical strength σ_1 predicted by Eq. (30). Consequently, the microcracks tend to open in regions where $\varphi_a<\varphi$, referred to as the microcrack opening zone. Oppositely, when φ_a is larger than φ, the local rock strength σ_{1a} is greater than the theoretical rock strength σ_1. The microcracks stabilize and tend to close in the zone of $\varphi_a>\varphi$, defined as the microcrack closing zone. Neglecting the interactions between microcracks, the stress state can be considered approximately equivalent within the local zones. Accordingly, the macroscopic rock strength is determined by the maximum strength of the microcrack clusters, corresponding to the most favorable orientation φ_m within the microcrack opening zone, calculated as Eq. (33).[25] Furthermore, the orientation of the macroscopic failure plane is controlled by this dominant microcrack orientation φ_m, representing the collective response of the microcrack clusters. Within this framework, the far-field principal stresses and the locally uniform stress state within the region serve as the effective boundary conditions for the microcrack cluster. To sum up, the microcrack cluster model is constructed by extending the analytical solution for a single penny-shaped microcrack using the most favorable orientation φ_m, without relying on the discrete sampling of microcracks.

$$\sigma_1=\sigma_{1a}(\varphi_m) \tag{33}$$

where σ_{1a} is the local rock strength corresponding to the actual rotation

angle of microcracks. φ_m represents the most favorable orientation of the microcrack cluster.

Substituting Eq. (33) into Eq. (30), the microfracture strength criterion extended to the microcrack cluster is expressed as

$$\sigma_1 - \sin^2\varphi_m\sigma_2 - \cos^2\varphi_m\sigma_3 = \sqrt{m_i\sigma_c\left(\sin^2\varphi_m\sigma_2 + \cos^2\varphi_m\sigma_3\right) + \sigma_c^2} \tag{34}$$

According to Eq. (34), for a constant σ_3, an increase in σ_2 leads to a corresponding rise in σ_1, reflecting the effect of σ_2. The φ_m, characterizing the macro-failure of rocks, closely correlates with stress state. To facilitate the handling of stress levels in the vector space, the stress state is represented through the Lode parameter γ.[55,56] An inherent characteristic and advantage of the ZZ strength criterion is its reduction to the HB strength criterion at both TC and TE states. To maintain consistency within this framework, the proposed microfracture strength criterion also satisfies this requirement. Therefore, based on the formulation of the ZZ strength criterion and the propagation of the microcrack cluster along the most favorable orientation φ_m, the influence of the stress state on φ_m within the microfracture strength criterion is concisely assumed by a linear correlation, with the specific derivation process is detailed below.

Combining Eq. (15) and Eq. (21), the propagation of the microcrack cluster along the most favorable orientation φ_m, is expressed as follows:

$$\begin{aligned}&(\sigma_1-\sigma_3)\left[\cos\theta\sin\theta\left(1-\gamma\sin^2\varphi_m\right)-\mu\gamma\sin^2\theta\sin^2\varphi_m\right]\\&-\mu\left(\sigma_1\cos^2\theta+\sigma_3\sin^2\theta\right)=\beta\end{aligned} \tag{35}$$

$$\begin{cases}A(\theta,\mu)(\sigma_1-\sigma_3)-B(\theta,\mu)(\sigma_1-\sigma_3)\gamma\sin^2\varphi_m-\mu\sigma_3=\beta\\A(\theta,\mu)=\cos\theta\sin\theta-\mu\cos^2\theta\\B(\theta,\mu)=\cos\theta\sin\theta+\mu\sin^2\theta\end{cases} \tag{36}$$

$$\frac{A(\theta,\mu)(\sigma_1-\sigma_3)-(\beta+\mu\sigma_3)}{B(\theta,\mu)(\sigma_1-\sigma_3)}\frac{1}{\gamma}=\sin^2\varphi_m \tag{37}$$

Since γ is a Lode parameter ranging between [0, 1], the fractional form of Eq. (37) requires further modification. Performing a Taylor expansion at points where $\gamma_0 \neq 0$ yields

$$\begin{aligned}\sin^2\varphi_m=\frac{A(\theta,\mu)(\sigma_1-\sigma_3)-(\beta+\mu\sigma_3)}{B(\theta,\mu)(\sigma_1-\sigma_3)}\Bigg[&\frac{1}{\gamma_0}-\frac{1}{\gamma_0^2}(\gamma-\gamma_0)+\frac{1}{\gamma_0^3}(\gamma-\gamma_0)^2\\&+\cdots+(-1)^n\frac{1}{\gamma_0^{n+1}}(\gamma-\gamma_0)^n\Bigg]\end{aligned} \tag{38}$$

Neglecting the higher-order terms and taking $\gamma_0 = 0.5$, Eq. (38) is simplified as

$$\sin^2\varphi_m \approx -4\frac{A(\theta,\mu)(\sigma_1-\sigma_3)-(\beta+\mu\sigma_3)}{B(\theta,\mu)(\sigma_1-\sigma_3)}(\gamma-1) \tag{39}$$

Accordingly, the correlation between $\sin^2\varphi_m$ and γ is reasonably approximated by a linear correlation in Eq. (40).

Table 1
Fitting results for 9 rocks.

Rock type	Rock	σ_c (MPa)	m_i[a]	C_1
Low m_i rock	Solnhofen limestone[34]	310	4	−0.3805
	TCDP siltstone[57]	64	9	−0.3735
	Mizuho trachyte[34]	100	10	−0.3545
	Yamaguchi marble[34]	82	12	−0.3600
Medium m_i rock	Coconino sandstone[33]	56.1	20	−0.3330
	Orikabe monzonite[34]	234	20	−0.3120
High m_i rock	KTB amphibolite[58]	165	30	−0.3065
	Inada granite[34]	229	30	−0.2970
	Westerly granite[28]	201	35	−0.2600

[a] Optimal m_i value fitted by the Zhang-Zhu strength criterion within empirical range recommended by Hoek and Brown,[59] Zhang and Zhu,[9] Marinos and Hoek.[60].

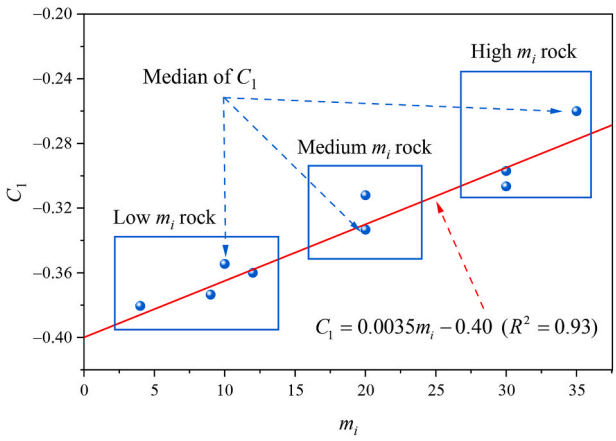


Fig. 4. Correlation between m_i and C_1 for 9 rocks.

$$\sin^2\varphi_m = C_1(\gamma-1) \tag{40}$$

$$\gamma=\frac{\sigma_2-\sigma_3}{\sigma_1-\sigma_3}=\frac{1-\sqrt{3}\tan\theta_\sigma}{2} \tag{41}$$

where C_1 is the variable determined by rock material, γ is the Lode parameter, and θ_σ is the Lode angle representing the stress state.

Combining Eq. (34) and Eq. (40), the microfracture strength criterion is rewritten as

$$\begin{aligned}&\sigma_1-(\gamma-1)C_1\sigma_2-(1+C_1-C_1\gamma)\sigma_3\\&=\sqrt{m_i\sigma_c[(\gamma-1)C_1\sigma_2+(1+C_1-C_1\gamma)\sigma_3]+\sigma_c^2}\end{aligned} \tag{42}$$

The C_1 is related to the intrinsic properties of rocks such as m_i. To further explore the correlation between C_1 and m_i, Eq. (42) is applied to fit true triaxial test data from 9 rocks. To minimize the influence of outliers or extreme values, the median of C_1 is adopted as the representative fitting result for each rock. The rocks are classified into low, medium, and high m_i rocks, and the correlation between C_1 and m_i is summarized in Table 1. The results indicate an increasing trend of C_1 with higher m_i. Furthermore, the fitted correlation in Fig. 4 demonstrates a linear relationship between C_1 and m_i, with a fitting accuracy (R^2) of 0.93.

$$C_1 = 0.0035m_i - 0.40 \quad (R^2 = 0.93) \tag{43}$$

Combining Eqs. (41)–(43), the microfracture strength criterion for the microcrack cluster is obtained.

$$\begin{cases}\sigma_1-\xi\sigma_2-(1-\xi)\sigma_3=\sqrt{m_i\sigma_c[\xi\sigma_2+(1-\xi)\sigma_3]+\sigma_c^2}\\\xi=\dfrac{-1-\sqrt{3}\tan\theta_\sigma}{2}(0.0035m_i-0.4)\end{cases} \tag{44}$$

Eq. (44) is reformulated in terms of $F(J_2, I_1, \theta_\sigma)$ as follows:

$$\begin{aligned}&\left[3\xi^2\sin^2\theta_\sigma+(2-\xi)^2\cos^2\theta_\sigma+2\sqrt{3}\,\xi(2-\xi)\sin\theta_\sigma\cos\theta_\sigma\right]J_2\\&+m_i\sigma_c\left[(1-\xi)\cos\theta_\sigma+\frac{\sqrt{3}}{3}(3\xi-1)\sin\theta_\sigma\right]\sqrt{J_2}-m_i\sigma_c\frac{I_1}{3}-\sigma_c^2=0\end{aligned} \tag{45}$$

where the relationship between the principal stresses σ_1, σ_2, σ_3 and I_1, J_2, θ_σ are defined below

$$\begin{bmatrix}\sigma_1\\\sigma_2\\\sigma_3\end{bmatrix}=-\frac{2}{\sqrt{3}}\sqrt{J_2}\begin{bmatrix}\sin\left(\theta_\sigma-\frac{2}{3}\pi\right)\\\sin(\theta_\sigma)\\\sin\left(\theta_\sigma+\frac{2}{3}\pi\right)\end{bmatrix}+\begin{bmatrix}I_1/3\\I_1/3\\I_1/3\end{bmatrix} \tag{46}$$

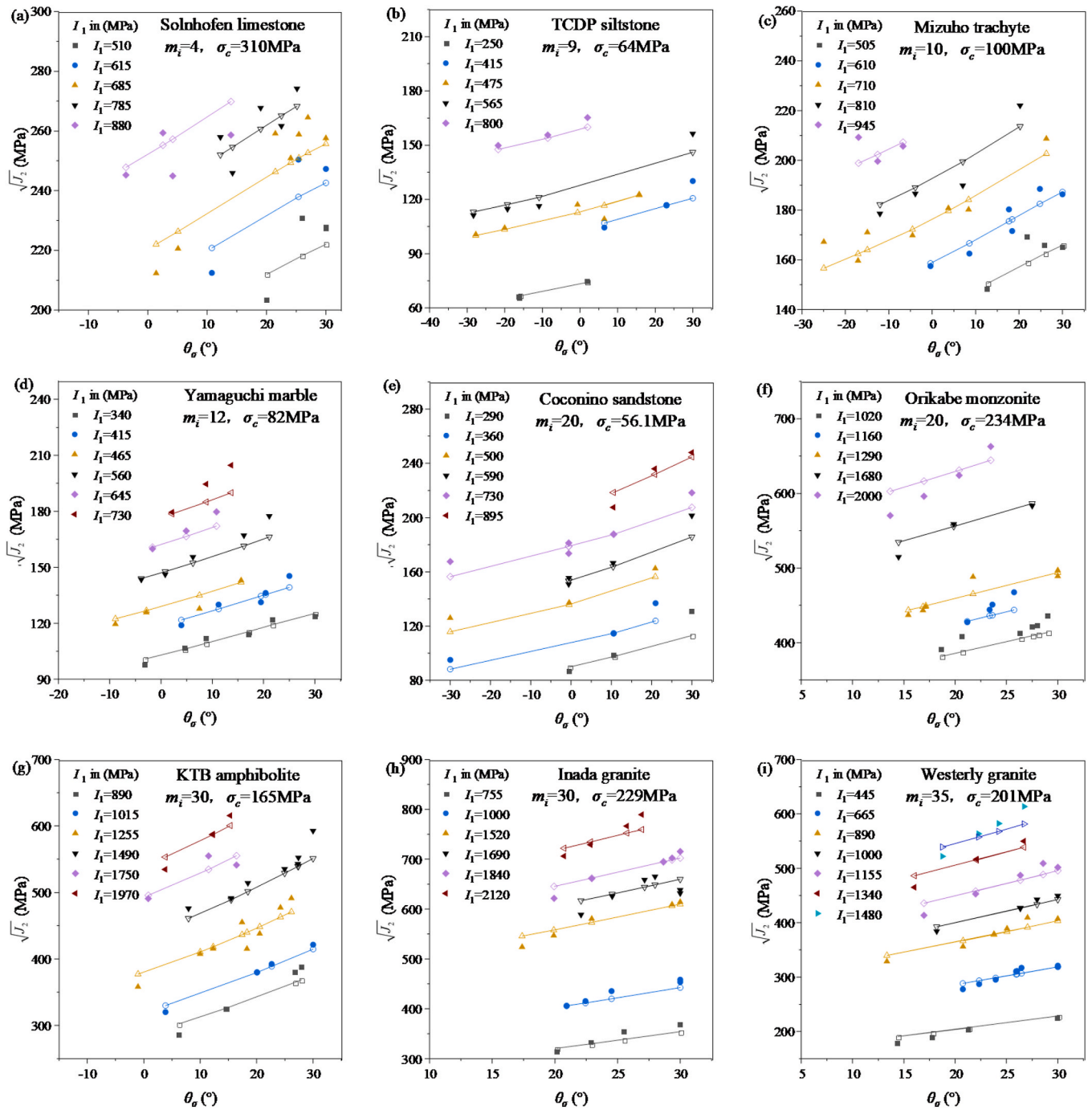


Fig. 5. Comparison of tested $\sqrt{J_2}$ (solid points) and theoretical $\sqrt{J_2}$ (hollow points) of micro-Zhang-Zhu strength criterion for 9 rocks in J_2-θ_σ space.

By combining Eq. (15) and Eq. (46), the mapping between the stress invariants and the microcrack stresses is expressed as follows:

The microfracture strength criterion reflects the nonlinear dependence of rock strength σ_1 on the principal stresses σ_2 and σ_3. For TC and

$$\begin{cases} \sigma_{11}' = \sqrt{J_2}\left[\sin\theta_\sigma\left(\dfrac{1}{\sqrt{3}} - \sqrt{3}\sin^2\theta\sin^2\varphi\right) + \cos\theta_\sigma\left(\cos^2\theta - \sin^2\theta\sin^2\varphi\right)\right] + \dfrac{I_1}{3} \\ \sigma_{12}' = \sqrt{J_2}\sin\theta\sin\varphi\cos\varphi\left(\cos\theta_\sigma - \sqrt{3}\sin\theta_\sigma\right) \\ \sigma_{13}' = \sqrt{J_2}\cos\theta\sin\theta\left[\sqrt{3}\sin\theta_\sigma\sin^2\varphi + \cos\theta_\sigma\left(1+\cos^2\varphi\right)\right] + \dfrac{I_1}{3} \end{cases} \tag{47}$$

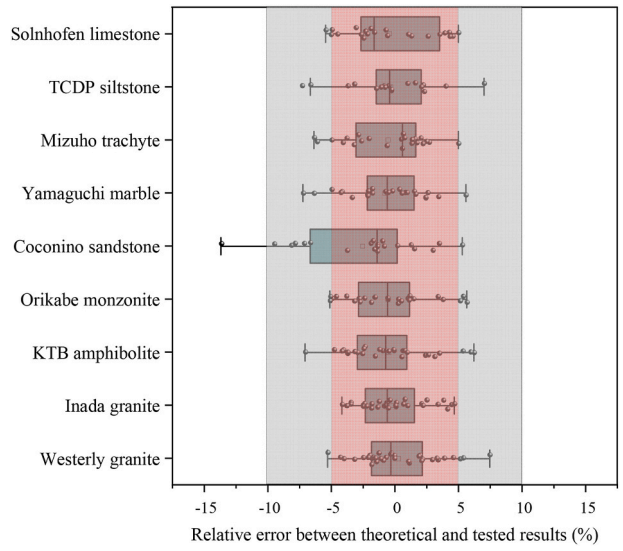


Fig. 6. Distribution of relative errors between tested results of true triaxial tests and theoretical results of micro-Zhang-Zhu strength criterion for 9 rocks.

TE states, it reduces to the theoretical HB strength criterion, which can be considered the true 3D HB strength criterion in the same way as the ZZ strength criterion. As it is developed from microfracture mechanics following the formulation of the ZZ strength criterion, it is herein defined as the micro-ZZ strength criterion.

3.4. Validation of micro-ZZ strength criterion

The effect of stress state on φ_m [Eq. (40)] and the correlation between C_1 and m_i [Eq. (43)] are assumed to be simple linear relationships in the derivation process, which requires further verification. The reasonability of the assumptions is assessed by verifying the accuracy of micro-ZZ criterion with true triaxial data for 9 rocks in Table 1. As a material with inherent defects, rock contains random microcracks within its structure, which is commonly regarded as approximately isotropic at the specimen scale. Therefore, the true triaxial test data are employed primarily to validate the capability of the micro-ZZ strength criterion in assessing rock strength at the macroscopic level. The theoretical $\sqrt{J_2}$ is

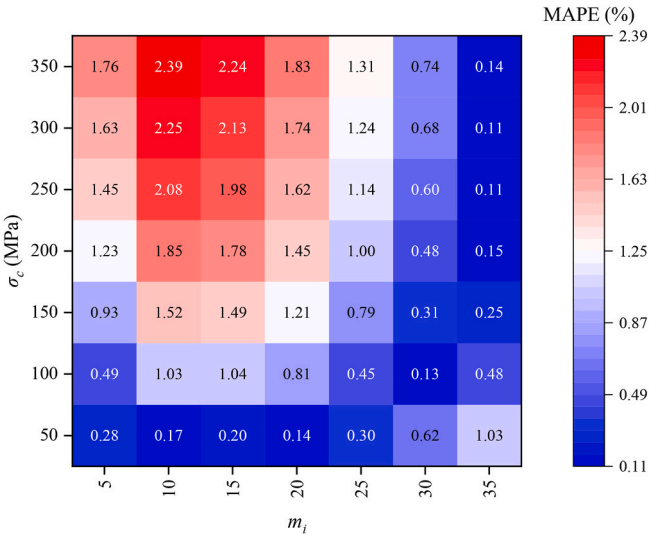


Fig. 8. Heatmap of mean absolute percentage error (MAPE) between Zhang-Zhu strength criterion and micro-Zhang-Zhu strength criterion.

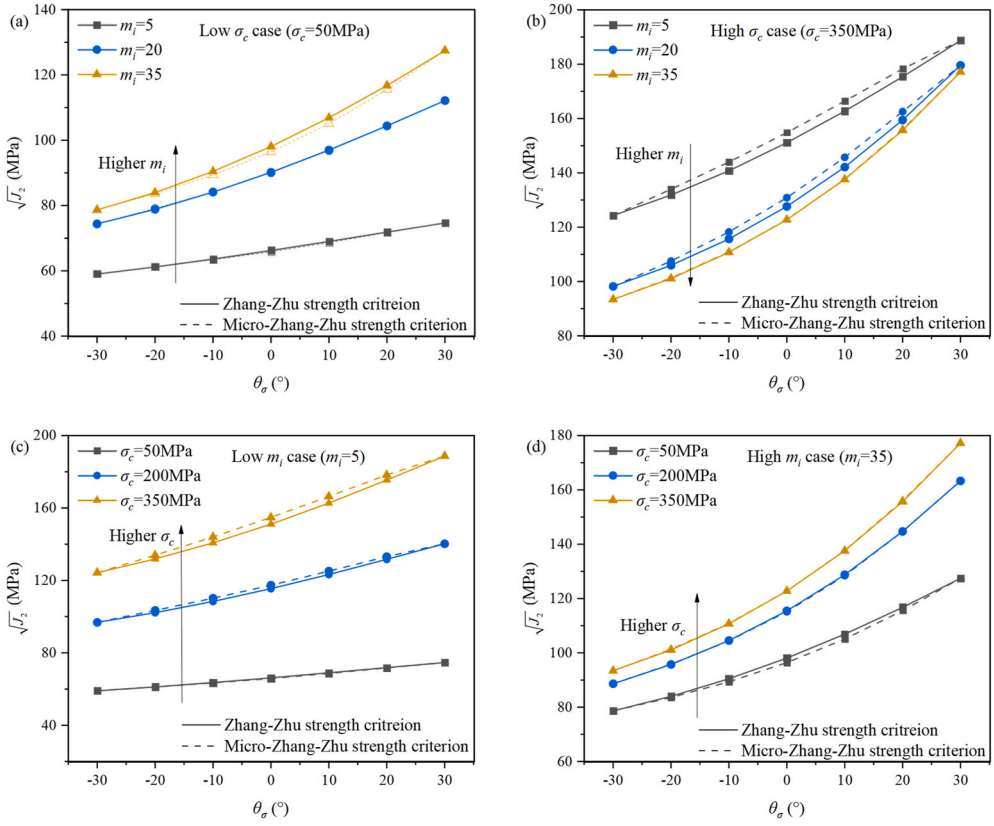


Fig. 7. Comparison of Zhang-Zhu strength criterion and micro-Zhang-Zhu strength criterion for different m_i and σ_c cases: (a) different m_i for low σ_c case; (b) different m_i for high σ_c case; (c) different σ_c for low m_i case; (d) different σ_c for high m_i case.

obtained by Eq. (45) and compared with the tested $\sqrt{J_2}$ for 9 rocks in J_2-θ_σ space, as illustrated in Fig. 5. The straight line and hollow points represent the theoretical $\sqrt{J_2}$, while the solid points represent the tested $\sqrt{J_2}$ from true triaxial tests. These results indicate that the theoretical $\sqrt{J_2}$ increases linearly with increasing θ_σ, which agrees well with the tested $\sqrt{J_2}$.

The relative errors between the theoretical and tested results are calculated to quantitatively assess the accuracy of the micro-ZZ strength criterion, as shown in the boxplot in Fig. 6. The mean and median relative errors for the 9 rocks are comparable, both falling within the range of ±2.5%. The Coconino sandstone displays considerable variability in the relative error, as evidenced by the extended box length and extreme outliers exceeding ±10%. Nevertheless, most data points for 9 rocks remain within the ±5% range. The statistical analysis reveals that 87.2% of the data points fall within a ±5% relative error, while 99.6% are within a ±10% relative error. This implies that the theoretical $\sqrt{J_2}$ closely matches the tested $\sqrt{J_2}$, indicating that the micro-ZZ strength criterion accurately predicts rock strength and that the linear assumptions underlying its derivation are valid. Moreover, the fracture behavior from microcrack clusters shows strong consistency with the true triaxial test results, thereby supporting the reliability of the proposed strength criterion.

To further assess the validity of the micro-ZZ strength criterion as a microfracture-based form of the ZZ strength criterion, Fig. 7 compares the theoretical $\sqrt{J_2}$ of both in rocks with different m_i and σ_c. The results show strong consistency between the two strength criteria, with their theoretical $\sqrt{J_2}$ exhibiting an approximately linear increase with the θ_σ. As m_i increases, the difference between the two strength criteria increases in the low σ_c case but decreases in the high σ_c case. Similarly, as σ_c increases, the difference enlarges for low m_i and narrows for high m_i. To quantitatively evaluate these differences, the mean absolute percentage error (MAPE) is employed and defined as

$$\mathrm{MAPE} = \frac{100\%}{n} \cdot \sum_{i=1}^{n} \left| \frac{\sqrt{J_{2Z,i}} - \sqrt{J_{2m,i}}}{\sqrt{J_{2Z,i}}} \right| \tag{48}$$

where n is the number of samples, $\sqrt{J_{2Z,i}}$ and $\sqrt{J_{2m,i}}$ represent $\sqrt{J_2}$ for the ZZ and micro-ZZ strength criteria for sample i.

The MAPE for m_i ranging from 5 to 35 and σ_c from 50 MPa to 350 MPa are calculated according to Eq. (48) and presented in Fig. 8. The MAPE is lower when both m_i and σ_c are extremely low or high, while larger MAPE is observed for low m_i and high σ_c. Although the MAPE is influenced by rock parameters, the maximum MAPE is only 2.39%, indicating that the micro-ZZ strength criterion closely aligns with the ZZ strength criterion.

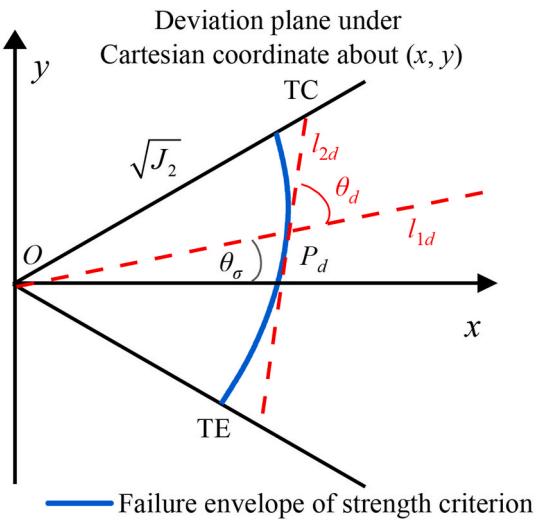


Fig. 10. Intersection angle θ_d for failure envelope of strength criterion in deviatoric plane.

4. Analysis of non-convexity for ZZ strength criterion based on microfracture mechanics

4.1. Interpretation of non-convexity for ZZ strength criterion

The non-convexity for the ZZ strength criterion in the deviatoric plane is further analyzed through the micro-ZZ strength criterion, providing an interpretation based on the underlying microfracture mechanisms. Fig. 9 illustrates the failure envelopes of the ZZ and micro-ZZ strength criteria in the deviatoric plane for low, medium, and high m_i cases. The results indicate that the two criteria are highly consistent, with both failure envelopes showing non-smoothness and non-convexity. This suggests that non-convexity is an intrinsic feature of failure envelopes derived from microfracture mechanics.

The failure envelope is transformed from a polar coordinate system about (J_2, θ_σ) to a Cartesian coordinate system about (x, y) to quantitatively represent the non-convexity in the deviatoric plane. Utilizing $x = \sqrt{J_2}\cos\theta_\sigma$ and $y = \sqrt{J_2}\sin\theta_\sigma$, the expressions of the ZZ and micro-ZZ strength criteria are converted as follows:

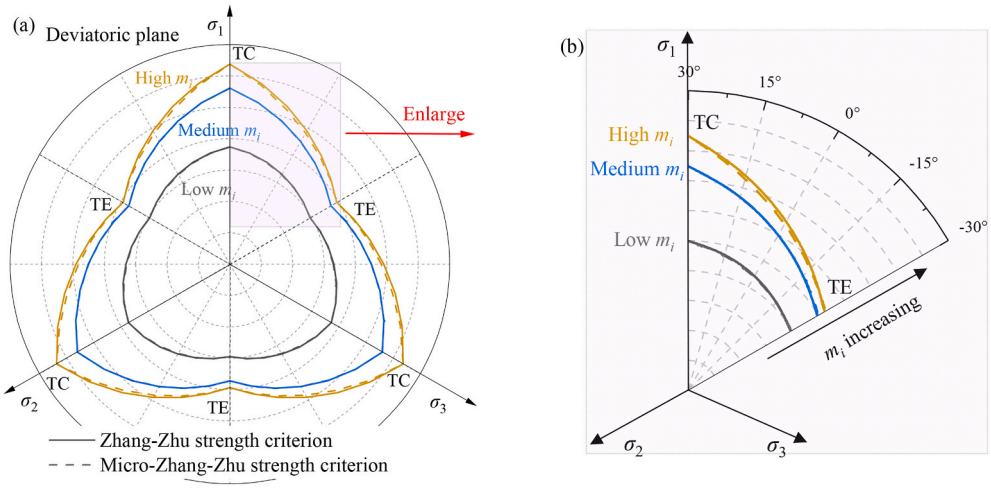


Fig. 9. Representation of Zhang-Zhu and micro-Zhang-Zhu strength criterion: (a) failure envelopes in deviatoric plane; (b) variation from triaxial compression to triaxial extension states.

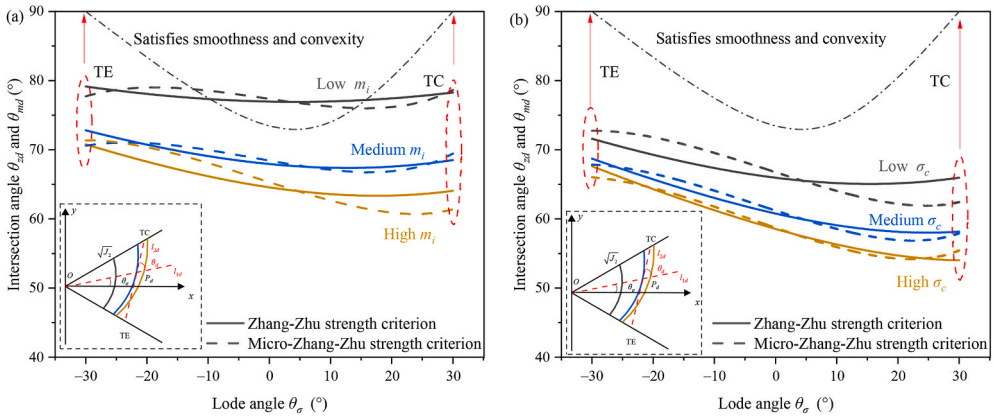


Fig. 11. Comparisons of θ_{md} for micro-Zhang-Zhu strength criterion and θ_{zd} for Zhang-Zhu strength criterion: (a) different m_i cases; (b) different σ_c cases.

$$\frac{3}{\sigma_c}(x^2+y^2)+\frac{\sqrt{3}m_i}{2}\sqrt{x^2+y^2}-\frac{m_i}{\sqrt{3}}y-\frac{m_i}{3}I_1-\sigma_c=0 \quad (\text{Zhang}-\text{Zhu}) \tag{49}$$

$$\left[\left(2+\frac{C_1}{2}\right)x-\frac{3}{2}C_1\frac{y^2}{x}\right]^2+m_i\sigma_c\left[\frac{2+C_1}{2}x-\frac{\sqrt{3}}{3}y-\frac{3C_1}{2}\frac{y^2}{x}\right] -m_i\sigma_c\frac{I_1}{3}-\sigma_c^2=0 \quad (\text{micro}-\text{Zhang}-\text{Zhu}) \tag{50}$$

Fig. 10 shows the failure envelope in the deviation plane under the Cartesian coordinate about (x, y). The point P_d represents the corresponding stress state and rock strength. OP_d is defined as l_{1d}, the distance from O to P_d is $\sqrt{J_2}$, and the angle between l_{1d} and the x-axis is θ_σ. The tangent to the failure envelope at P_d is denoted as l_{2d}, and the angle formed by l_{1d} and l_{2d} is defined as the intersection angle θ_d. Let the tangential slopes of l_{1d} and l_{2d} be k_{1d} and k_{2d}, the intersection angle θ_d is calculated as follows:

$$\begin{cases}\tan\theta_d=\dfrac{k_{2d}-k_{1d}}{1+k_{1d}k_{2d}}\\ k_{1d}=\tan\theta_\sigma\\ k_{2d}=\dfrac{\partial y}{\partial x}\end{cases} \tag{51}$$

According to Eqs. (49)–(51), the slope k_{2d} corresponding to the failure envelope is determined to obtain the specific expression for the intersection angle θ_d.

For the ZZ strength criterion, the intersection angle θ_{zd} is expressed as

$$\tan\theta_{zd}=\frac{12\sqrt{3}\cdot\sqrt{J_2}+3m_i\sigma_c-2m_i\sigma_c\sin\theta_\sigma}{2m_i\sigma_c\cos\theta_\sigma} \tag{52}$$

For the micro-ZZ strength criterion, the intersection angle θ_{md} is expressed as

$$\begin{cases}\tan\theta_{md}=\dfrac{t_1-t_2\tan\theta_\sigma}{t_2+t_1\tan\theta_\sigma}\\ t_1=\left[(4+C_1)^2x^2-\dfrac{9C_1^2y^4}{x^2}\right]\dfrac{1}{2x}+m_i\sigma_c\left(\dfrac{2+C_1}{2}+\dfrac{3C_1y^2}{2x^2}\right)\\ t_2=\left[(4+C_1)x-\dfrac{3C_1y^2}{x}\right]\dfrac{3C_1y}{x}+m_i\sigma_c\left(\dfrac{\sqrt{3}}{3}+\dfrac{3C_1y}{x}\right)\\ C_1=0.0035m_i-0.40\end{cases} \tag{53}$$

Based on Eq. (52) and Eq. (53), the intersection angle θ_d for the Lode angle θ_σ ranging from −30° to 30° is calculated for different m_i and σ_c cases, as illustrated in Fig. 11. It is found that the θ_{zd} for the ZZ strength criterion closely aligns with the θ_{md} for the micro-ZZ strength criterion. As the Lode angle θ_σ increases, the θ_{zd} tends to decrease and then increase, whereas the θ_{md} fluctuates around the θ_{zd} without reaching 90° at either the TC or TE states. This means that both failure envelopes neither smooth or convex. The comparisons of the intersection angle θ_d for different m_i and σ_c reveal a decreasing trend in θ_d with higher m_i and σ_c, which manifests as a more pronounced non-convex failure envelope. Meanwhile, the difference between the θ_{zd} and θ_{md} increases with higher m_i and decreases with larger σ_c. In summary, the non-convexity for micro-ZZ strength criterion, as reflected by the failure envelope and intersection angle θ_{md}, confirms that one of the reasons for the non-convexity for the ZZ strength criterion is the microcrack mechanism of microcrack clusters in rocks, thereby establishing a correlation between microfracture mechanisms and the macroscopic non-convexity inherent in the Zhang-Zhu strength criterion.

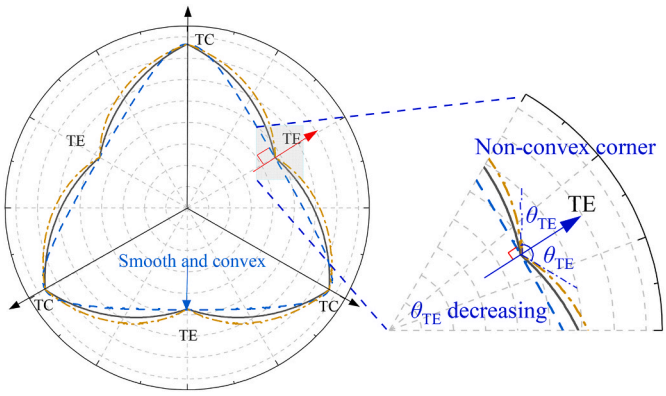


Fig. 12. Correlation between non-convex corner θ_{TE} and the shape of failure envelope in deviatoric plane.

4.2. Physical mechanism of non-convexity for ZZ strength criterion

The above analysis indicates that the non-convexity for the ZZ strength criterion is attributed to the microfracture mechanism of the microcrack cluster. Nevertheless, how the microcrack cluster influences the degree and evolution of non-convexity warrants further investigation. The underlying physical mechanism of the non-convexity for ZZ strength criterion is revealed by exploring the effect of microcrack density within the rock on the non-convexity.

Since the non-convexity in the deviatoric plane manifests as a concave failure envelope at the TE state, the non-convex corner (θ_{TE}) serves as a quantitative index of non-convexity, defined as the intersection angle at the TE state. The non-convex corners θ_{zTE} and θ_{mTE} are obtained as special solutions of Eq. (52) and Eq. (53) at $\theta_\sigma=-30°$.

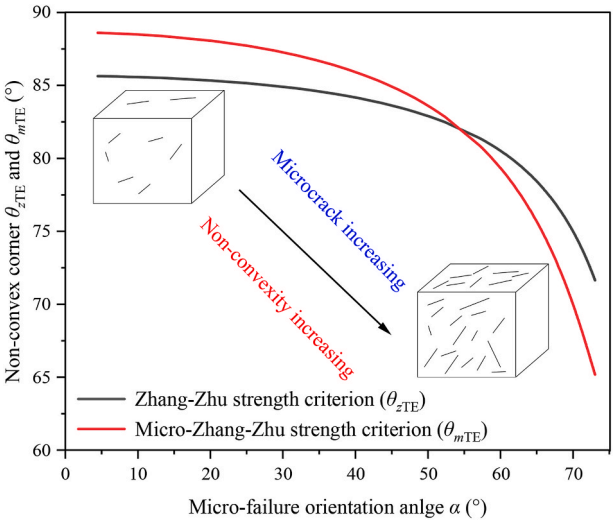


Fig. 13. Correlation between non-convex corners and micro-failure orientation angle α

$$\begin{cases} \theta_{zTE} = \theta_{zd,\theta_\sigma=-30^\circ} \\ \theta_{mTE} = \theta_{md,\theta_\sigma=-30^\circ} \\ m_i = \dfrac{2\mu\sigma_c}{\kappa\sigma_t(2-\nu)} \\ \kappa = \sqrt{\dfrac{3(1-\nu)}{(2+2\nu-\nu^2)}} \end{cases} \tag{54}$$

where the θ_{zTE} and θ_{mTE} represent the non-convex corners for the ZZ and micro-ZZ strength criteria at TE states, respectively. A larger non-convex corner corresponds to less non-convexity, while the failure envelope becomes smooth and convex when the non-convex corner reaches 90°, as illustrated in Fig. 12. Since the micro-ZZ strength criterion satisfy that failure occurs once the distortional strain energy exceeds a critical threshold, the κ is determined as $\sqrt{3(1-\nu)/(2+2\nu-\nu^2)}$ by Sih and Macdonald.[61]

The microfracture density is positively correlated to α, thereby allowing α to serve as an index of microcrack density. At the TE state, the α is expressed as a function of $\sqrt{J_2}$ and I_1.

$$\cos\alpha = \frac{\mu}{\sqrt{1+\mu^2}}\left(-\frac{1}{3}+\frac{2I_1}{3\sqrt{3}\sqrt{J_2}}+\frac{\kappa\sigma_t(2-\nu)}{\mu\sqrt{J_2}}\right) \tag{55}$$

According to Eq. (32), m_i is determined by the parameters μ, κ, σ_c/σ_t, and ν. Maintaining these parameters constant ensures a constant m_i, which means that the analyzed object is microscopically the same type of rock. The microcrack density, represented by α, and the non-convexity, quantified by θ_{TE}, are not independent causal variables, which are governed by the stress state and material parameters (μ, ν, σ_c/σ_t, σ_c, and I_1). Their relationship is described by an implicit form $F(\theta_{TE}, \alpha; \mu, \nu, \sigma_c/\sigma_t, \sigma_c, I_1) = 0$ rather than an explicit functional form $\theta_{TE} = f(\alpha)$. Variations in α arise from changes in these parameters, which simultaneously induce corresponding variations in θ_{TE}. Therefore, the relationship between θ_{TE} and α corresponds to an implicit curve that reflects the parametric dependence, as illustrated in Fig. 13. The micro-ZZ strength criterion shows larger variation in the non-convex corner than the ZZ strength criterion, mainly because the former is based on microfracture mechanics, while the latter is established through empirical fitting. However, the non-convex corners of both criteria remain relatively close and exhibit similar overall trends, which effectively reflect the inherent non-convex characteristics of the ZZ strength criterion.

Considering the intact structure of real rocks, microcracks are sparsely distributed and unlikely to reach high microcrack densities. Therefore, the correlation between the non-convexity and microcrack density shown in Fig. 13 is investigated under the premise of low microcrack densities. The non-convex corner shows a decreasing trend with higher α, indicating that higher microcrack density contributes to greater non-convexity in the ZZ strength criterion. This trend becomes pronounced as α increases, reflecting the growing influence of microcracks concentrated within the crack propagation zone on the macroscopic failure envelope. Although microcrack interactions may become non-negligible as microcrack density increases, their effect on the overall trend remains limited. From Fig. 13, when the micro-failure orientation angle α is small, the non-convex corner approaches 90°, particularly for the micro-ZZ strength criterion. Extrapolation of the current results shows that for rocks without microcracks, the non-convex corner converges to 90°, corresponding to a convex failure envelope consistent with classical criteria (such as von Mises or Drucker-Prager) for homogeneous or uncracked materials. This suggests that the presence of microcracks within the rock is one of the micro-mechanical mechanisms contributing to the non-convexity for the ZZ strength criterion. Moreover, an increase in microcracks leads to a more pronounced manifestation of the non-convexity. This does not imply that the non-convexity originates solely from the microcracks. Other factors, including mineralogical heterogeneity, grain interlocking, and anisotropy, may also influence the non-convexity, which are smaller-scale mechanisms that warrant further investigation in future studies.

The non-convexity of the ZZ strength criterion is an intrinsic characteristic of rock failure. In numerical calculations, enforcing artificial convexity through smoothing may obscure strength weakening regions and overestimate the rock strength, particularly near the TE state. Therefore, retaining the non-convexity in the numerical schemes based on the ZZ strength criterion enables more accurate strength evaluation, with corresponding implementation approaches recently proposed.[62]

5. Discussion

5.1. Effect of stress state and material parameters on non-convexity

From Fig. 11, the non-convexities for the ZZ and micro-ZZ strength criteria are affected by rock parameters such as m_i and σ_c. The rock parameter m_i, as an empirical parameter of the HB strength criterion, is determined by Eq. (32). To highlight the microfracture mechanisms of the non-convexity in the deviatoric plane, the effect of stress state and material parameters on the non-convexity is explored. The non-convex corner θ_{zTE} and θ_{mTE} determined in Eq. (52) serve as quantitative indices of non-convexity, which is influenced by the five parameters, including the first principal stress invariant I_1, the compression-extension stress ratio σ_c/σ_t, the uniaxial compressive strength σ_c, the Poisson's ratio ν, and the friction coefficient μ. For common rocks, the σ_c/σ_t typically ranges from 5 to 35, the σ_c ranges from 50 MPa to 350 MPa, and the I_1 ranges from 100 MPa to 700 MPa. According to Gercek,[63] the Poisson's ratio ν for rock typically ranges from 0.1 to 0.4. The friction coefficient μ is between 0.15 and 1.35 based on the internal friction angle of common rocks.[64,65]

Fig. 14 illustrates the effects of the stress state and material parameters on the non-convex corner of the ZZ and micro-ZZ strength criteria. As the uniaxial compressive strength σ_c increases, both θ_{zTE} and θ_{mTE} decrease simultaneously, and the rate of decrease gradually slows down. Conversely, the first principal stress invariant I_1 shows a positive correlation with the non-convex corner for both strength criteria. For the increment of the compression-extension stress ratio σ_c/σ_t, the θ_{zTE} tends to decrease, while the θ_{mTE} decreases and then increases, especially for larger σ_c. According to Fig. 14(e) and (f), for the ZZ strength criterion, the non-convex corner θ_{zTE} is negatively linearly correlated with the Poisson's ratio ν, and decreases as μ increases. As for the micro-ZZ strength criterion, the correlation between θ_{mTE} and ν gradually changes from a negative linear correlation to a positive linear

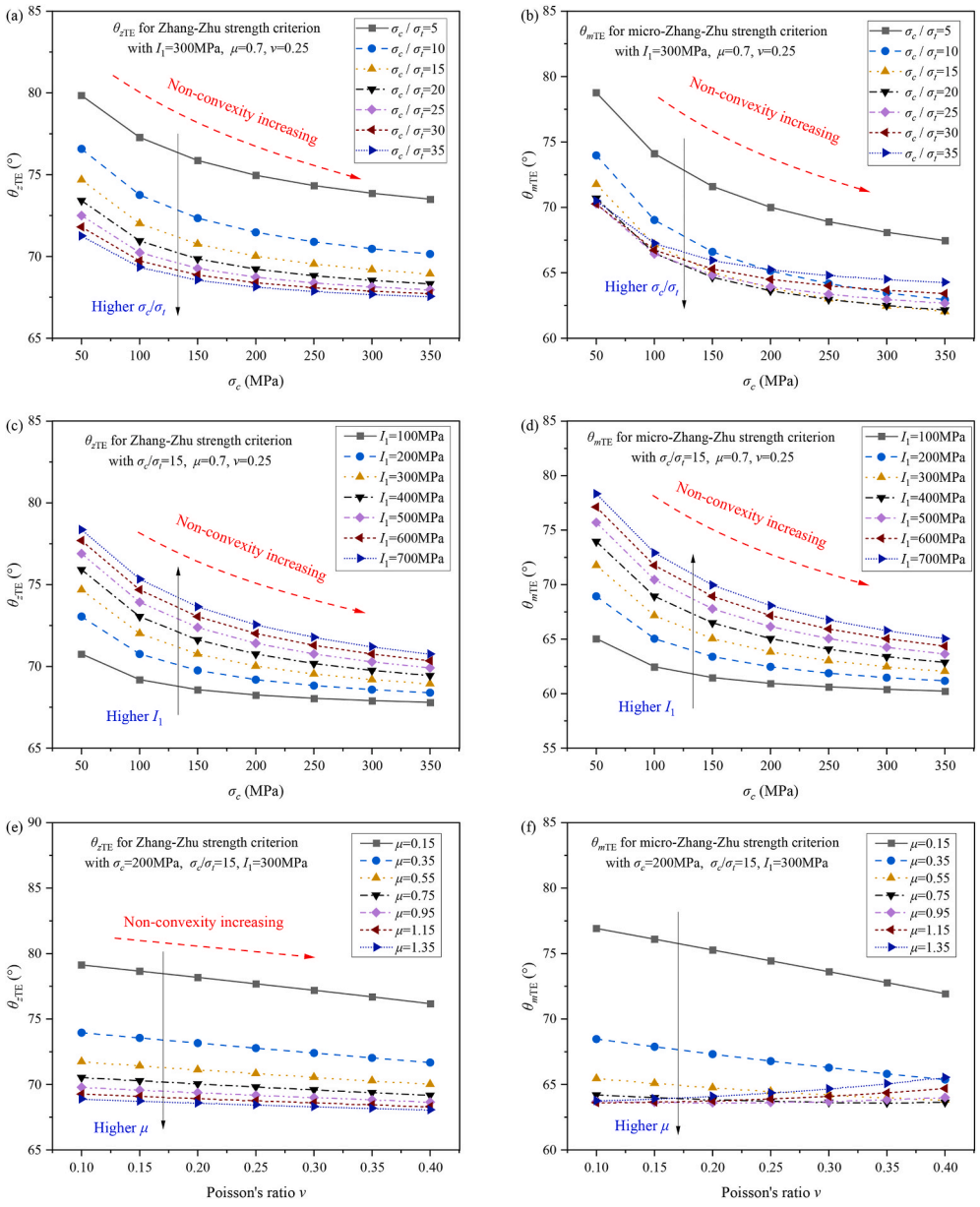


Fig. 14. Effects of stress state and material parameters on non-convex corners for Zhang-Zhu strength criterion (θ_{zTE}) and micro-Zhang-Zhu strength criterion (θ_{mTE}): (a) θ_{zTE} for different σ_c and σ_c/σ_t; (b) θ_{mTE} for different σ_c and σ_c/σ_t; (c) θ_{zTE} for different σ_c and I_1; (d) θ_{mTE} for different σ_c and I_1; (e) θ_{zTE} for different μ and ν; (f) θ_{mTE} for different μ and ν

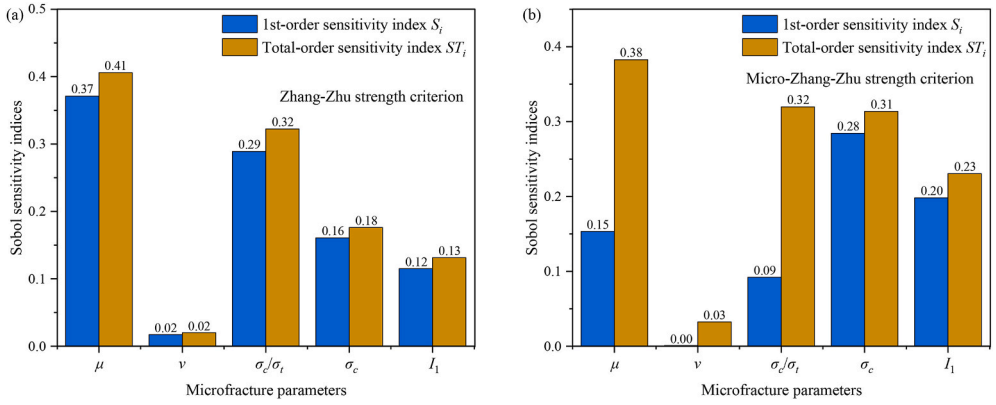


Fig. 15. 1st-order (S_i) and total-order (ST_i) sensitivity indices for Zhang-Zhu strength criterion and micro-Zhang-Zhu strength criterion: (a) S_i and ST_i for Zhang-Zhu strength criterion; (b) S_i and ST_i for micro-Zhang-Zhu strength criterion.

correlation as μ increases. Meanwhile, the correlation between $\theta_{m\mathrm{TE}}$ and μ changes from a positive correlation to decrease and then increase after $\nu > 0.15$, with the increasing trend becoming more pronounced at higher ν. The effects of σ_c/σ_t, μ, and ν on the non-convexity for the micro-ZZ strength criterion shift from a positive to a negative correlation as their values increase. This trend indicates that higher values of these parameters tend to suppress the propagation of microcracks within the rock. Such microstructural effects, however, are not reflected by the original ZZ strength criterion.

5.2. Sensitivity analysis for non-convexity

To quantify the extent of the effect of the stress state and material parameters on the non-convexity, sensitivity analyses are performed for both the ZZ and micro-ZZ strength criteria. The Sobol sensitivity analysis (SSA) is widely employed to evaluate the relative contribution (in terms of variance) of each input and the interactions to the model output.[66–70] The contribution of inputs and their interactions on the model output is expressed as

$$\mathrm{Var}(Y) = \sum_i V_i + \sum_{i<j} V_{ij} + \sum_{i<j<k} V_{ijk} + \cdots + V_{1,2,\ldots,n} \tag{56}$$

where $Y = f(X)$ is the output of the model, $\mathrm{Var}(Y)$ denotes the total variance of the model output, $V_i = \mathrm{Var}(E(Y|X_i))$ is the contribution of a single input X_i, V_{ij} denotes the interaction between X_i and X_j, and the $V_{1,2,3,\ \ldots,n}$ represents the high-order interaction effects for all input parameters.

To assess these contributions, the Sobol sensitivity index is defined by Sobol[67] as follows:

$$\begin{cases} S_i = \dfrac{V_i}{\mathrm{Var}(Y)} \\ S_{ij} = \dfrac{V_{ij}}{\mathrm{Var}(Y)} \\ ST_i = \dfrac{V_i + \sum_{j\neq i} V_{ij} + \sum_{j\neq i,k\neq i} V_{ijk} + \cdots}{\mathrm{Var}(Y)} \end{cases} \tag{57}$$

where S_i is the 1st-order sensitivity index indicating the main contribution of the X_i to the output, ignoring interactions with other parameters. S_{ij} is the 2nd-order sensitivity index representing the contribution of the interaction between X_i and X_j to the output. ST_i is the total-order sensitivity index reflecting the overall contribution of the X_i, including the interaction with other parameters.

The Sobol sensitivity analysis is applied to assess the effect of five parameters on the non-convexity, where the input parameters are μ (0.15~0.35), ν (0.1~0.4), σ_c/σ_t (5~35), σ_c (50~350), and I_1 (100~700), and the output are $\theta_{z\mathrm{TE}}$ and $\theta_{m\mathrm{TE}}$. Based on the range of five input parameters, 5000 samples are generated using the Monte Carlo method, and the Sobol sensitivity indices are calculated, as presented in Fig. 15. Neglecting interactions among input parameters, the 1st-order sensitivity indices S_i indicate that the friction coefficient μ exerts the largest influence on the non-convexity for the ZZ strength criterion, while the most influential parameter for the micro-ZZ strength criterion is σ_c. Considering the interactions among input parameters, the total-order sensitivity index ST_i indicates that the friction coefficient μ exerts the greatest influence on the non-convexity for both strength criteria, while the Poisson's ratio ν contributes the least. Although the S_i of the ZZ and micro-ZZ strength criteria show differences in the independent effects of the parameters on non-convexity, the ST_i of both strength criteria reflect similar parameter effects after considering the interactions among input parameters. The micro-ZZ strength criterion exhibits stronger parameter interactions than the ZZ criterion, due to its basis in microfracture mechanics, which better reflects the intrinsic correlations among stress state and material parameters.

The 2nd-order sensitivity index S_{ij} is calculated to further explore the interactions between the stress state and material parameters, as shown in Fig. 16. Although, the S_{ij} for the micro-ZZ strength criterion is nearly 10 times larger than that for the ZZ strength criterion, the relative magnitudes of the interactions between the input parameters are consistent. Both indicate the strongest interaction between the friction coefficient μ and the compression-extension stress ratio σ_c/σ_t. The high interaction between the two input parameters stems mainly from Eq. (32). As for the other input parameters, the Poisson's ratio ν interacts most strongly with the friction coefficient μ, and the uniaxial compressive strength σ_c interacts most significantly with the first principal stress invariant I_1. In summary, the interaction between μ and σ_c/σ_t is the strongest, and these two parameters exert significant effects on the non-convexity for the ZZ strength criterion. Since μ and σ_c/σ_t dominate the non-convexity, accurate characterization of the friction coefficient and the tensile strength is essential for reliably characterizing the shape of the failure envelope.

6. Conclusion

This study explores the non-convexity for the ZZ strength criterion in

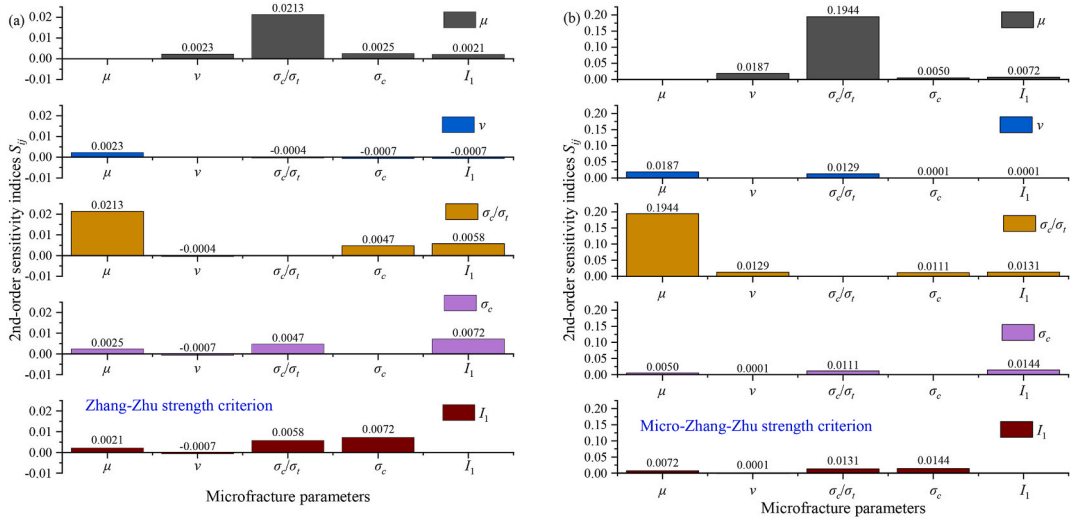


Fig. 16. 2nd-order sensitivity indices S_{ij} for Zhang-Zhu strength criterion and micro-Zhang-Zhu strength criterion: (a) S_{ij} for Zhang-Zhu strength criterion; (b) S_{ij} for micro-Zhang-Zhu strength criterion.

the deviatoric plane from microfracture mechanics and analyzes the effect of the parameters on the non-convexity. The conclusions are as follows:

(1) The ZZ strength criterion shows intrinsic non-convexity in the deviatoric plane, as evidenced by the true triaxial test data from Coconino sandstone ($I_1 = 740$ MPa $\pm$ 5%) aligning with its failure envelope. This implies the non-convexity is correlated to the underlying rock failure mechanisms.
(2) A micro-ZZ strength criterion is derived from microfracture mechanics, following the formulation of the ZZ strength criterion. Validation against the true triaxial test reveals that fracture behavior from microcrack clusters agrees well with true triaxial test results, with 87.2% of the data points falling within a $\pm$5% relative error. Compared to the ZZ strength criterion, the micro-ZZ strength criterion shows a maximum MAPE of only 2.39%, confirming its high consistency with the ZZ strength criterion.
(3) In the deviatoric plane, the failure envelope of the micro-ZZ strength criterion closely coincides with that of the ZZ strength criterion, with a high degree of agreement in their intersection angles θ_d. This correspondence establishes a correlation between microfracture mechanisms and the macroscopic non-convexity inherent in the Zhang-Zhu strength criterion. Furthermore, the non-convexity corner approaches 90° as the microcrack density tends to 0, indicating that the presence of microcracks within the rock constitutes one of the micromechanical mechanisms underlying the non-convexity of the ZZ strength criterion, with more microcracks resulting in more pronounced non-convexity.
(4) For the ZZ strength criterion, the non-convex corner shows a positive correlation with the I_1 and negative correlations with the σ_c, σ_c/σ_t, ν, and μ. In contrast, for the micro-ZZ strength criterion, the effects of σ_c/σ_t, μ, and ν on the non-convexity shifts from positive to negative as their values increase, indicating that higher values of these parameters suppress the microcrack propagation. Sobol sensitivity analysis reveals that μ and σ_c/σ_t are dominant parameters affecting non-convexity, with the strongest interaction between them.

For compressive and shear-dominated rock failure under true triaxial stress states, Mode II is generally considered primary, while the potential contributions of additional fracture modes warrant further investigation. Moreover, this study adopts a simplified model that neglects the interactions between microcracks, which may affect rock strength and the non-convexity of the failure envelope. Incorporating such interactions in future work could provide a more accurate understanding of the microscopic mechanisms governing the non-convexity of the ZZ strength criterion.

CRediT authorship contribution statement

Qi Zhang: Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Yixin Shen:** Writing – original draft, Software, Methodology, Investigation. **Hehua Zhu:** Supervision, Project administration. **Xiaojun Wang:** Validation, Data curation. **Yuechao Pei:** Visualization, Formal analysis.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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